

CONTINUATION.md — vasic-digital tmux

Last updated: 2026-05-28T15:00Z

§0 — How to resume work in any CLI agent

Paste this prompt:

Read CONTINUATION.md at the repo root. Identify the topmost item under §3 Active work with status IN PROGRESS or BLOCKED. Re-read Constitution.md, CLAUDE.md, AGENTS.md, Issues.md, Fixed.md for mandates and current backlog. Resume from current state. Update this document as you work.

§1 — Snapshot

Field	Value
Repo	vasic-digital/tmux on GitHub + GitLab
Origin	Migrated from ATMOSphere project (scripts/tmux/, docker/Dockerfile.tmux-build, docs/guides/TMUX_OPTIMIZED_BUILD.md) on 2026-05-07
Pinned tmux	upstream tag 3.6a
Version	1.0.14 (versionCode 15) — A35 clipboard copy-OUT physical proof (test 44 + M44 + TMUX-CH-44 + verify.sh Layer-1 extension) + A36 e2e stale-podman-prereq fix + B3 P5 escape transparency; multi-host deploy (Mistborn + nezha), 2026-05-22
Verification (this cycle)	Multi-host deploy GREEN. Mistborn (Darwin arm64): bash scripts/setup.sh --rebuild → GREEN; suite PASS=41 FAIL=0 SKIP=3. NEW test 44 PASS=7/0/0 with T5 pbpaste returning the marker — physical end-user clipboard proof. e2e PASS=9/0/0 (after A36 fix). Meta-test 39 caught / 2 escaped / 8 skipped (escapes pre-existing P5-M20/M21, see Issues.md B3). Nezha (Linux ALT 6.12 x86_64): bash scripts/setup.sh --rebuild → GREEN; suite PASS=37 FAIL=0 SKIP=7 (SKIP 44 because T5 honestly SKIPS on headless server with no DISPLAY/Wayland; T1-T4 binding-chain proof all PASS — exactly the §104 topology dispatch the test was designed for). e2e PASS=9/0/0 including T7 distinct cgroup scopes for two operator-path sessions. Meta-test 37 caught / 4 escaped / 8 skipped (P5-M20/M21 same pre-existing + M22 environmental-

Field	Value
Governance docs	CodeGraph-state issue, see B3 — neither introduced by v1.0.14). Captured 2026-05-22 on both hosts. constitution/ submodule (HelixConstitution, pinned 84c948d); Containers/ submodule (pinned fbef9d6); project Constitution.md (Project Articles §101–§109, extends the submodule), CLAUDE.md, AGENTS.md, QWEN.md, Issues.md, Fixed.md, this document

§2 — Mandates (canonical authority)

- Constitution.md §1 anti-bluff covenant (verbatim user-mandate quote propagated)
- Constitution.md §11.4.1 FAIL-bluffs equally forbidden
- Constitution.md §11.4.2 recorded-evidence requirement
- Constitution.md §11.4.3 per-host-topology test dispatch
- Constitution.md §11.4.4 test-interrupt-on-discovery + 4-layer test coverage
- Constitution.md §11.4.5 audio + video quality analysis (N/A for tmux scope; principle generalized)
- Constitution.md §11.4.6 — No-guessing mandate (verbatim user-mandate quote)
- Constitution.md §9 absolute data safety
- Constitution.md §12.6 60% host memory budget (per-session TMX_MEM is the enforcement)
- Constitution.md §5 / §12.10 continuation-document sacred invariant

§3 — Active work

§3.1 Governance bring-up — Issues.md / Fixed.md / explicit §11.4.x anchors landed

Status: COMPLETE (2026-05-08T17:06Z).

- Created Issues.md (open / in-flight tracker) with:
 - Categories A-E, status conventions OPEN / PARTIAL / BLOCKED / RUNNING / INVESTIGATED
 - Status reclassification rules referencing §11.4.6
 - Seeded items: META-MUT-001, CHAL-COVER-001, TEST-AUDIT-001, TMX-T5, TMX-T7, TMX-T8, TOPO-DISPATCH-001
- Created Fixed.md (closed archive) with:
 - A0 initial migration (commit 08d4ba5, 2026-05-07)
 - B0 §1 covenant propagation (commit b92bf7f, 2026-05-08)
 - C0 test 09 crash-isolation-scope landed and green (commit b92bf7f, 2026-05-08)
- Updated Constitution.md:
 - Explicit §11.4.1, §11.4.2, §11.4.3, §11.4.4, §11.4.5 anchors added (preserving §1 prose verbatim)
 - §11.4.6 no-guessing mandate added with verbatim user-mandate quote
 - §9 absolute data safety added
 - §12.6 60% host memory budget added (TMX_MEM enforcement seam)

- §5 expanded into full §12.10 continuation-document mandate
- Updated CLAUDE.md and AGENTS.md to carry every numbered anchor + canonical-authority cross-references

§3.2 Per-session containerization (Phase B — multi-session work)

Status: RESEARCH COMPLETE + WRAPPER LANDED + ISOLATION VERIFIED (2026-05-08).

Web research output: docs/plans/containerization.md recommends systemd-run --user --scope (cgroup-v2 transient scope) over podman-per-session. Wrapper at scripts/tmx implements tmx {new|attach|ls|kill} with MemoryMax=\$TMX_MEM (default 8G), CPUQuota=\$TMX_CPU (default 200%), TasksMax=4096, Delegate=yes.

Test 09 (scripts/tests/09_crash_isolation_scope.sh) — 14 PASS / 0 FAIL / 0 SKIP — verifies T1 (host capability), T2 (wrapper invariants), T3 (cgroup interface evidence), T4 (SIGKILL containment + user.slice survival), T6 (concurrent registration independence). Source-line breakdown:

- **T1:** systemd 258 + cgroup v2 mounted ✓
- **T2:** tmx wrapper invokes systemd-run --user --scope + sets MemoryMax/CPUQuota/TasksMax/Delegate=yes ✓
- **T3:** transient scope created; /sys/fs/cgroup/.../memory.max reads 268435456 bytes (matches set 256M) + /sys/fs/cgroup/.../cpu.max reads 50000 100000 (50% quota over 100ms period) — both POSITIVE EVIDENCE per §1 ✓
- **T4:** spawn scope, read MainPID from cgroup.procs, SIGKILL it, verify scope inactive AFTER kill, verify default.target=active THROUGHOUT (user.slice survives — Constitution §1 invariant) ✓
- **T6:** 3 concurrent scopes, all registered + active simultaneously ✓

§3.3 Pending tests — moved to Issues.md

The previously-listed deferred tests (T5 memory pressure under cap, TasksMax stress, Concurrent OOM independence) have migrated to Issues.md:

- **TMX-T5** — Issues.md C1 (PARTIAL; operator-unblock runbook for dedicated test host)
- **TMX-T7** — Issues.md C2 (OPEN; new test 11 planned)
- **TMX-T8** — Issues.md C3 (OPEN; new test 12 planned)

Cross-cutting blockers also tracked in Issues.md:

- **META-MUT-001** — Issues.md A1 (paired-mutation harness needed before §11.4.4 layer-4 coverage is real)
- **CHAL-COVER-001** — Issues.md B1 (HelixQA Challenge entries pending)
- **TOPO-DISPATCH-001** — Issues.md D1 (formal topology dispatch matrix)

§3.4 Anti-bluff enforcement — test audit + challenges fix + covenant propagation verification

Status: COMPLETE (2026-05-08T18:00Z).

- Verified anti-bluff covenant propagation across ALL governance files:
 - Root Constitution.md: §1 + §11.4.1-§11.4.6 present ✓
 - Root CLAUDE.md: full covenant present ✓
 - Root AGENTS.md: compact covenant present ✓
 - Containers/CLAUDE.md: full covenant present ✓
 - Containers/AGENTS.md: full covenant present ✓
 - tmux/ (upstream submodule pinned to 3.6a): N/A — no governance files, not modified per Constitution
- Performed line-by-line §11.4.2 anti-bluff audit of tests 01-09:
 - All 9 tests confirmed to carry positive runtime evidence (/proc files, cgroup interfaces, tmux command output, systemctl state).
 - No FAIL-bluff vectors found (all set -uo pipefail, guarded variables).
 - Full audit table moved to Fixed.md B2.
- Fixed broken paths in scripts/challenges/tmux.yaml (scripts/tmux/tests/ → scripts/tests/).
- Migrated TEST-AUDIT-001 from Issues.md B2 → Fixed.md B2.

§3.5 Hostname-derived status-bar colour — algorithm + wrapper + 2 tests + challenges

Status: COMPLETE (2026-05-08T19:00Z).

- Created scripts/hostname_color.sh — DJB2 hash over hostname → curated 27-colour palette. Deterministic, testable standalone.
- Updated scripts/tmux.template — added _apply_host_color() function that invokes hostname_color.sh and applies set -g status-style bg=<colour> on the running tmux server. Applied on both new-session and attach paths.
- Created scripts/tests/10_hostname_color_algorithm.sh — 5 invariants (deterministic, valid colourNNN, palette member, spread ≥12/16, empty-fallback to system hostname). All 5 PASS on this host.
- Created scripts/tests/11_hostname_color_integration.sh — verifies wrapper applies correct colour to tmux server. SKIPs if binary/wrapper not yet built.
- Updated scripts/tests/run_all.sh — glob changed from 0[1-9]*.sh to [0-9][0-9]*.sh to support tests 10+.
- Updated scripts/challenges/tmux.yaml — added TMUX-CH-10 (algorithm) and TMUX-CH-11 (integration).
- Updated docs/research/customization/colors.md — added implementation documentation covering algorithm, integration, verification, usage, and anti-bluff covenant.
- Updated AGENTS.md — bump test count from 9 to 11.

§3.6 Full coverage — missing challenge entries, destructive tests, topology dispatch

Status: COMPLETE (2026-05-08T20:00Z).

- **CH-09:** Added missing challenge entry for crash isolation scope (test 09) — was the only test without a challenge.
- **Test 12** (12_memory_pressure_under_cap.sh): TMX-T5 — allocates up to MemoryMax+10% inside a transient scope, captures dmesg OOM-kill evidence, validates user.slice survival. Gated by TMX_TEST_DESTRUCTIVE=1.
- **Test 13** (13_tasksmax_stress.sh): TMX-T7 — fork-bomb resistance. Spawns processes up to TasksMax=4096, reads pids.current/pids.max from cgroup interface. Gated by TMX_TEST_DESTRUCTIVE=1.
- **Test 14** (14_concurrent_oom_independence.sh): TMX-T8 — three scopes A/B/C, OOM-kills A, verifies B and C survive with original MainPIDs. Gated by TMX_TEST_DESTRUCTIVE=1.
- **CH-12/13/14:** Challenge entries for all three destructive tests.
- **Topology dispatch:** added _probe_topology() to scripts/tmx.template — detects systemd version + cgroup v2, classifies host as tmx-supported/tmx-degraded/tmx-unsupported per §11.4.3.
- **Fixed.md B2:** closure SHA updated to 68a65b0.
- **Issues.md → Fixed.md migration:** B1 (CHAL-COVER-001), C1 (TMX-T5), C2 (TMX-T7), C3 (TMX-T8), D1 (TOPO-DISPATCH-001) moved to Fixed.md.

§3.8 Audit cycle 2026-05-13 — submodule pin-drift caught + governance staleness fixed

Status: COMPLETE (2026-05-13T00:00Z).

Triggering event: user invoked /init followed by "what is left unfinished, no-bluff policy heavily enforced everywhere". Audit found and remediated:

- **CRITICAL §1 / §11.4.6 bluff:** f4132aa Auto-commit (2026-05-13) silently bumped tmux submodule from cc117b5 (tag 3.6a) to 3f651d9f (3.6a-329-g3f651d9f — 329 commits past the only stable tag). Governance docs across Constitution / CLAUDE / AGENTS / CONTINUATION / README still claimed "pinned to tag 3.6a". User decision: revert to tag (newer stable tag does not exist; 329 commits are unreleased upstream master). Submodule pointer reverted to cc117b5.
- **README.md staleness:** line 5 ("eight verification tests"), line 33 (PASS=6 FAIL=0 SKIP=2), line 37 ("8 tests cover"), line 39 (2-SKIP list) — all rewritten to current 14-test / PASS=10 SKIP=4 / 4-SKIP-list state with explicit TMX_TEST_DESTRUCTIVE=1 callout and §11.4.4 layer-4 harness reference.
- **CLAUDE.md drift:** added top-of-file project summary + canonical-doc cross-links + fresh-conversation workflow; fixed 0N_*.sh glob → NN_*.sh; named the four layers inline in the Test-interrupt rule; added "Files to never edit directly" section (parity with AGENTS.md).
- **AGENTS.md drift:** mirrored CLAUDE.md project summary + cross-links; fixed same 0N_*.sh glob bug; added meta-test row to commands table.
- **Issues.md format:** restored A / B / C / D / E section headers as empty sections (was: A / B / E only, with C and D missing despite conventions list referencing them). All bodies note "landed in Fixed.md" with item IDs for traceability.

- **CONTINUATION.md timestamp:** 2026-05-08T22:00Z → 2026-05-13T00:00Z; this entry added under §3.8 per §12.10 invariant.
- **M6 mutation added** (scripts/tests/meta_test_false_positive_proof.sh): injects \$\$ into the hostname_color hash, making the algorithm non-deterministic per-invocation. Test 10 T1 must FAIL under this mutation. Closes the audit-flagged gap: until M6, the "same-host = same-color" user invariant was protected by side-effect only (no dedicated anti-randomness mutation).
- **docs/guide/README.md phantom-script bluff fixed:** 3 occurrences each of verify_tmux.sh → verify.sh, setup_tmux.sh → setup.sh, install_tmux_deps.sh → install_deps.sh; "8 tests" → "14 tests" (2×).
- **GUIDE.md "severity hierarchy" pre-existing bluff caught** (Fixed.md A3): documented "blockers / critical / advisory" classification does not exist in run_all.sh — every test is treated equally (any FAIL = RED). Rewritten to describe the actual gate logic. Caught while extending the table for tests 09-14; would have been silently doubled otherwise.
- **Verification + validation cycle 2026-05-13** (Fixed.md A14): operator invoked superpowers:verification-before-completion + asked for full anti-bluff verification + governance propagation. Every verification command run FRESH this session per Iron Law. Captured evidence: setup.sh --verify-only SUMMARY=PASS=10 FAIL=0 SKIP=5 GREEN; test_e2e.sh PASS=9/0/0 GREEN; test 15 PASS=6/0/0; live tmx new -s VerifyDemo shows operator's host shell (milosvasic@Mistborn, which brew=/opt/homebrew/bin/brew, ulimit -t=86400, ulimit -u=2666). Two defects caught and fixed: (D1) wrapper hostname resolution drifted to FQDN after bridge removal — restored scutil-LocalHostName path so colour stays colour202 for Mistborn; (D2) install_deps.sh + build_oom_set.sh weren't OS-aware out of the box — now install_deps.sh detects Darwin → brew install (no sudo), build_oom_set.sh SKIPS on non-Linux. Governance propagation completed: verbatim user-mandate quote now in root CLAUDE.md + AGENTS.md (previously only in Constitution + Containers); §11.4.7 reference added to Containers/CLAUDE.md + Containers/AGENTS.md (previously only Containers/CONSTITUTION.md). Install state: shell snippets in both ~/.bashrc + ~/.zshrc, zsh -c 'which tmx; tmx -V' returns native wrapper + tmux 3.6a. macOS RLIMIT_AS gap documented honestly per §1 (XNU EINVAL for unprivileged setrlimit on memory).
- **Per-session isolation + Constitution §11.4.7** (Fixed.md A13): user reported core@localhost + lack of OOM containment. Forensic: tmx new -s isol1 -d + tmx new -s isol2 -d previously shared ONE cgroup (run-p504653-i504654.scope) — README's "if one session OOMs, others survive" was a §1 bluff. Architectural rewrite: each tmx new -s NAME now spawns its OWN tmux server on socket tmx-NAME inside its OWN scope tmx-NAME.scope with host-adaptive MemoryMax (max(MemTotal × 60% / 4, 2GB)), CPUQuota=200%, TasksMax=4096, Delegate=yes. OOM in any session = contained to that scope; B+C survive with original MainPIDs (verified test 14 PASS=8/0/0 with stress-ng tmx send-keys). Test 15 NEW: per-session cgroup distinctness (6 assertions, positive readback from /sys/fs/cgroup). Test 11 + test 14 + test 08 rewritten to OPERATOR PATH (tmx new -s NAME instead of hand-crafted systemd-run --user --scope). e2e T7 NEW: bridge confirms two distinct scopes via VM systemctl. Meta-test M5 retargeted, M7 retargeted, M9+M10 NEW (10/10 mutations caught). Plan + operator decisions captured in docs/plans/per-session-isolation.md. **Constitution §11.4.7 NEW** (operator-path test coverage rule): every gate test MUST exercise the same entry point an end-user invokes; tests that hand-craft equivalents are supplementary.

Propagated to Containers/CONSTITUTION.md at same depth. Forensic anchor: Fixed.md A12 (status-bar green default) + A13 (sessions sharing one cgroup) — both shipped while non-operator-path tests reported GREEN.

- **Regression protection for A10 — gate-coverage gap closed** (Fixed.md A11): user demanded "make sure nothing passes again! zero-bluff policy MUST BE followed blindly!" — honest accounting of why the existing gates missed the colour bug:
 - test 11 always passed `-S "$SOCKET"` → tested explicit-socket path only; operator's default-socket use case uncovered.
 - meta-test M4/M5 targeted `scripts/tmx` (the SSH bridge on Darwin) instead of `scripts/tmx-vm` (the actual VM wrapper) — paired-mutation harness was mutating the wrong file.
 - test `e2e.sh` pre-fix didn't read `status-style` at all.
 - Triple-layer regression protection added: (1) test 11 T6 — new test exercising default-socket path with positive evidence (`tmux show -g status-style` with NO `-S`, FAILs explicitly on `bg=green` default-applied bluff); (2) meta-test M4/M5 retargeted to `scripts/tmx-vm`; (3) M7 (re-introduce SOCK-empty early return) + M8 (hardcode `bg=green`) both caught by test 11 T4.1/T6. Result: all 8 mutations caught + reverted (16 PASS), full destructive suite PASS=14/0/0, `e2e` PASS=8/0/0.
- **Status-bar colour silently green (default) + bridge ignored macOS hostname** (Fixed.md A10): user said "Bottom was green for current host. Is that expected for mistborn?" — two stacked bluffs caught:
 - `_apply_host_color / _apply_oom_score` had `[ -n "$sock" ] || return 0` — silently bailed when `tmx new -s` Test was invoked WITHOUT `-S` (which is the primary operator use case; `tmux` uses its default socket then). So `set -g status-style` was never fired → `tmux` default `bg=green` shown. Test 11 always passes `-S "$SOCKET"` so the bug was uncovered. Fix: helpers now build `target=(-S "$sock")` only when `sock` is non-empty; otherwise omit `-S` and let `tmux` use its default socket.
 - Bridge `scripts/tmx-mac.template` didn't forward macOS hostname → `hostname_color.sh` inside VM resolved against `localhost.localdomain` regardless of macOS host. Every macOS user got the same VM-derived colour, violating host-distinguishability. Fix: bridge captures `scutil --get LocalHostName` (e.g., "Mistborn"), forwards as `TMX_HOSTNAME=<host>` in the remote SSH command. Wrapper's `_apply_host_color` passes `${TMX_HOSTNAME:+"$TMX_HOSTNAME"}` to `hostname_color.sh` — env var is source of truth when set, `$(hostname)` fallback otherwise.
 - New test `e2e.sh` T4.5 explicitly reads `tmx show -g status-style`, computes expected colour from `scutil --get LocalHostName` (or `hostname` on Linux), FAILs on `bg=green` (default-applied bluff) or non-match. Captured runtime evidence on Mistborn: `status-bg 'colour202'` matches hostname-derived 'colour202' for 'Mistborn'.
- **Interactive `tmx new fixed` + `e2e` automation harness** (Fixed.md A9):
 - User ran `tmx new -s` Test interactively after install and hit open terminal failed: not a terminal. The bridge + `ssh -t` allocated TTY correctly; the WRAPPER's `cmd & wait` pattern was disconnecting `tmux` from the foreground process group → no TTY access. Detached-mode tests (`-d`) all passed because `tmux` daemonizes; interactive (no `-d`) hit the bluff. §11.4.1 FAIL-bluff in the wrapper itself.
 - Fix: wrapper now detects `-d/-D` in args (INTERACTIVE flag), and for interactive: runs `systemd-run` in foreground inheriting TTY, injects `-d` after

- the new-session keyword (so tmux server detaches), then `exec attach` to take over TTY interactively. For detached: keeps existing path.
- New: `scripts/test_e2e.sh` — 7-test operator-facing automation: prerequisites + `tmx -V` (via bridge) + new-session + send-keys + capture-pane + survive-without-client + kill-session. All with positive runtime evidence; trap-on-EXIT cleanup. Captured the literal marker string echoed inside the session pane through the bridge — proves the full operator stack carries intent end-to-end.
- Verified: `bash scripts/test_e2e.sh GREEN 7/0/0`;
`TMX_TEST_DESTRUCTIVE=1 bash scripts/test_vm.sh still GREEN 14/0/0` (wrapper change didn't regress existing tests).
- **macOS tmx fully operational + side-by-side end-to-end** (Fixed.md A8):
 - `bash scripts/setup.sh` on Darwin now diverges intelligently: builds (containerized), generates BOTH `scripts/tmx-vm` (Linux wrapper for VM paths) AND `scripts/tmx` (Darwin SSH bridge from new `scripts/tmx-mac.template`); runs `scripts/test_vm.sh` for §11.4 target-env verification (since binary cannot exec on Darwin); on GREEN, appends `bashrc` snippet to BOTH `~/.bashrc` AND `~/.zshrc` (macOS defaults to `zsh`).
 - **Bridge mechanics:** `scripts/tmx-mac.template` discovers `podman` machine's SSH endpoint via `podman machine inspect` at every call (port re-discovery — survives `podman machine stop && start`), verifies `tmx-vm` is executable in the VM, then `ssh -t -i <identity> -p <port> core@127.0.0.1 "<vm-repo>/scripts/tmx-vm <quoted-args>"`.
 - **Verified end-to-end on Darwin 24.5.0 / Apple Silicon:** `zsh -c 'source ~/.zshrc; which tmx; tmx -V' → /Users/.../scripts/tmx + tmux 3.6a`; `system /opt/homebrew/bin/tmux -V → tmux 3.6a` — side-by-side coexistence proven; VM-side verify GREEN PASS=11 FAIL=0 SKIP=3.
 - **scripts/verify.sh now respects \$WRAPPER and \$TMUX_BIN env vars** so `test_vm.sh` / bridge tooling can redirect to the right wrapper without source modifications.
 - **README.md** gained a new "Architecture" section with ASCII diagram showing Linux-native vs Darwin-bridge dispatch; `docs/guide/README.md` §5.5 documents the bridge mechanics + Mermaid flowchart; CLAUDE/AGENTS one-liners updated.
- **Final sweep — env-specific wrapper + §255 violations + sed portability** (Fixed.md A7):
 - **Environment-bound wrapper:** `scripts/tmx` was generated for one env (container / repo paths OR VM /Users/.../ paths) and silently failed in the other. New `scripts/test_vm.sh` orchestrator regenerates the wrapper with VM-native paths before each VM-side run. Both `test_containerized.sh` and `test_vm.sh` now own the wrapper-regeneration step for their respective target.
 - **Constitution §255 violations:** "ATMOSphere" in `GUIDE.md` title + body, challenge `yaml` description, `oom_set.c` license comment, `tmux.conf.template` attribution; `atm_tmux_test_*` / `atm_test_*` / `atm_tmux_test_color_*` socket/session names across 8 test files; `~/.tmux.conf.pre-atmosphere` backup naming. All renamed: code uses `tmx_*` / `pre-vasic-digital`; docs branded "vasic-digital".
 - **sed -i portability:** `setup.sh` used GNU-only `sed -i` for `bashrc` markers — would fail on macOS BSD `sed`. Centralized in `_strip_bashrc_snippet()` using `perl -i -ne ... .. / ... flip-flop range delete` (portable GNU+BSD).

- **Verified post-fix in VM:** `TMX_TEST_DESTRUCTIVE=1 bash scripts/test_vm.sh` → `PASS=14 FAIL=0 SKIP=0; META=1 bash scripts/test_vm.sh` → 12/12 mutations caught.
- **Install-mechanism side-by-side bluffs** (Fixed.md A6): user asked about the alias and side-by-side coexistence requirement. Audit found three real defects:
 - `bashrc` snippet pointed `PATH` at `scripts/tmux/` (phantom subdir; wrapper is at `scripts/tmx`).
 - `alias tmux='tmx'` line shadowed the system `tmux` command → broke side-by-side.
 - `setup.sh` closing message said "Then 'tmux' will use the ATMOSphere build" (wrong command name + stale branding).
 - **Canonical answer to user's question:** the alias is **tmx** (not `tmux`). Wrapper at `scripts/tmx` generated from `tmx.template`. `tmx new|attach|ls|kill` invokes the verified vasic-digital `tmux`; `tmux` invokes whatever was on the operator's `PATH` before (system / Homebrew / apt / dnf). Both coexist.
- **Full destructive + meta-test cycle caught 6 more FAIL-bluffs** (Fixed.md A5): pushing past `PASS=10 SKIP=4` into actually running `TMX_TEST_DESTRUCTIVE=1` + meta-test in the podman machine VM (Fedora CoreOS 42 + `systemd 257` + `stress-ng` + `tmx-oom-set` setcap) surfaced multiple §11.4.1 FAIL-bluffs. Final: **PASS=14 FAIL=0 SKIP=0** (full verify) + **12/12 mutations caught** (meta-test). Real defects fixed:
 - Tests 12/14 `_skip;<continue>` bug — printed `SKIP` but didn't exit; continued running and FAILED on missing `stress-ng` → false §11.4.1 FAIL-bluff. Fix: explicit `exit 0`.
 - Tests 12/14 unprivileged `dmesg` → fallback to `journalctl -k` via `_kring_count` / `_kring_tail` helpers; broader OOM regex.
 - Test 13 fork-storm OOM-killed its own scope (4096 sleeps × 700KB > 256M MemoryMax); two-phase scope, polling registration, test `TASKS_MAX 4096` → 256 with `MemoryMax=512M` (production wrapper still `TasksMax=4096` — separately grep-verified by test 09 T2.2).
 - Meta-test `sed -i strip-exec-bit` → `orig_mode` capture/restore via `stat` + `chmod`.
 - Meta-test M5 `sed` pattern was eval-expanded (`${TMX_MEM:-8G}` → 8G) so it never matched → silent no-op → mutation escaped (a **bluff in the gate itself**). Fix: simplified to literal `MemoryMax=` | `MemMax=` pattern.
 - Meta-test had no debug visibility → added opt-in `TMX_META_DEBUG=1`.
 - **The user's "same-host = same-color" invariant is now PROVEN end-to-end** with `tmux show -g status-style reading colour166` on first session AND on second session on same host (test 11 T4.1 + T5).
- **Build pipeline bluffs caught while reproducing on macOS** (Fixed.md A4): three real defects surfaced when user asked why we couldn't build on macOS:
 - `docker/Dockerfile`: `groupadd -g 20` collided with Ubuntu's `diabolo` group → build aborted at step 7/10. Fix: `-o` (non-unique) on `groupadd` + `useradd`. Build now reproducible on Darwin hosts.
 - `docker/build_inside_container.sh`: `LDFLAGS -ljemalloc` was being dropped by linker's default `--as-needed` because `tmux` doesn't reference `jemalloc` symbols. **README's "Build-time -ljemalloc" claim was a §1 bluff until this commit**. Fix: `-Wl,--no-as-needed -ljemalloc -Wl,--as-needed`. Verified `ldd` now shows `libjemalloc.so.2` in `DT_NEEDED`.
 - `docker/build_inside_container.sh`: `make -j2` silently no-op'd after `LDFLAGS` change (existing binary defeated mtime check). Fix: prepend `make clean` so `LDFLAGS` changes always take effect.

The f4132aa Auto-commit itself is a §12.10 / §11.4.6 violation — opaque commit message, silent submodule mutation, no CONTINUATION update. Cannot rewrite history per §9 data safety; documenting here is the audit trail.

§3.7 §11.4.4 layer-4 paired-mutation harness landed (A1 META-MUT-001)

Status: COMPLETE (2026-05-08T21:00Z).

- Created scripts/tests/meta_test_false_positive_proof.sh — §11.4.4 layer-4 harness with 5 registered mutations:
 - **M1**: break hostname_color output format → test 10 T2 FAILs (invalid format)
 - **M2**: force hash to zero → test 10 T3 FAILs (zero spread)
 - **M3**: single-entry palette → test 10 T3 FAILs (palette mismatch)
 - **M4**: remove systemd-run flag from tmx.template → test 09 T2 FAILs
 - **M5**: remove Delegate=yes from tmx.template → test 09 T2 FAILs
- Results on this host: **10 PASS / 0 FAIL / 0 SKIP** — all mutations caught.
- Updated CLAUDE.md: removed "PENDING" references, wired mandatory operations to the actual file.
- Updated AGENTS.md: layer 4 now marked as landed, not PENDING.
- Updated Fixed.md: A1 entry with closure details; all "PENDING META-MUT-001" references replaced with actual coverage references.
- Updated Issues.md: A1 removed (→ Fixed.md A1). Issues.md now has **zero open items** — all originally seeded items are closed.
- Cleaned up CONTINUATION.md §3.6: removed stale "remaining open" note.

§3.16 — v1.0.15 multi-line + paste-IN + alt-screen Claude Code + workable-items SQLite + DOCX + constitution sync (2026-05-28)

Status: IN PROGRESS — Mistborn build GREEN + Mistborn full suite GREEN + meta-test GREEN; nezha deploy pending; release tag pending.

Triggering event: operator mandate (verbatim, 2026-05-28):

"In last iteration of work we did we have allegedly fixed copy / paste from and to the tmux session window. Selecting multiple lines and copying of them does not work. We MUST BE able to scroll vertically everywhere and copy / paste anything! Especially in Claude Code (claude command)! Do in depth investigation and systematic debugging, reproduce issue and then apply comprehensive fixes! ... Every bug / issue we report MUST BE first covered with all supported tests which will actually confirm failure / problems we report! Then, after we apply fixes, the same tests MUST CONFIRM all changes and fixes really working! We MUST for every issue or change, for any workable item create proper workable item in our Issues doc and all relevant docs around it (with them all being exported in all expected file types - PDF, HTML, DOCX)! We MUST NEVER forget the flow: workable item → SQLite database → all docs we have related to workable items!"

Five-PWU pipeline executed per §11.4.58 + §11.4.94 (parallel-by- default + zero-idle). Phase 0 fetched + ff-merged constitution/ from 84c948d → 6828ff2 (19 commits) and Containers/ from fbef9d6 → 2e9ca0e (17 commits). PWU-A

(tmux conf + 4 tests + 4-layer coverage) + PWU-B (§11.4.87..98 propagation) + PWU-C (Go binary FULL Phase 3+ project-local) + PWU-D (DOCX export) ran in parallel; PWU-E (commit + deploy) is the closing sequential PWU.

- **A37 — Multi-line + PASTE-INTO + alt-screen TUI mouse-drag override** (Fixed.md A37): the user's primary ask. Identified the three uncovered surfaces vs v1.0.14: (1) multi-line drag in Claude-Code-like TUIs (mouse_any_flag=1 forwards drag to app), (2) paste-INTO from OS clipboard (set-clipboard external is copy-OUT only), (3) the alt-screen + mouse-tracking surface never asserted end-to-end. Added M-/S-MouseDrag1Pane overrides + matching drag-end @clip-routed binds + @clip-read user-option + prefix+P paste bind. Tests 45/46/47/48 land 4-layer coverage with pbpaste-back physical evidence: PASS=6/0/0 + 6/0/0 + 8/0/0 + 9/0/0 = 29 PASS, RED→GREEN proven for tests 46 + 48. Synthetic alt-screen surrogate synthetic_alt_screen_app.py substitutes for Claude Code in tests (no OAuth flake per §11.4.98). Mutations M46 + M48 CAUGHT + RESTORED.
- **A38 — Constitution + Containers sync + §11.4.87..98 propagation** (Fixed.md A38): submodule pointers advanced via --ff-only; short- form mirrors for 12 new universal anchors landed in project Constitution.md / CLAUDE.md / AGENTS.md / QWEN.md (each anchor appearing ≥3 times in each consumer per PWU-B audit). New Layer-1 gate L1C in verify.sh enforces literal presence.
- **A39 — Workable-items SQLite SSoT (Go binary, project-local Phase 3+)** (Fixed.md A39): constitution scaffold ships Phase-2 stubs only; project added cmd/workable-items/ with full Phase 3+ logic (sync md-to-db / sync db-to-md / diff / validate / add / close / report). 11 Go sources + embedded schema (drift-check-cited from constitution 6828ff2) + 10 tests passing 30/30 (3 iters per §11.4.50). Pure-Go modernc.org/sqlite (no CGO). Initial DB at docs/workable_items.db seeded with 45 items (ATM-001..ATM-045) from live Issues.md + Fixed.md. TRACKED in git per §11.4.95. Honest gaps logged: (a) legacy items default to Type=Task (no Type lines pre-§11.4.16); (b) live-corpus round-trip not byte- identical for free-form bodies (per §11.4.93 phase 6); (c) upstream PR + integration into commit_all.sh deferred.
- **A40 — DOCX export** (Fixed.md A40): scripts/sync_all_markdown_exports.sh extended to emit .docx siblings alongside .html + .pdf via pandoc. 44/44 candidates produced valid DOCX. Layer-1 gate CM-DOCX-EXPORT-SYNC enforces canonical-doc sibling presence.

Multi-host deploy + release plan (PWU-E, sequential):

1. ✓ Mistborn setup.sh --rebuild GREEN.
2. ✓ Mistborn full suite via verify.sh — **PASS=45 FAIL=0 SKIP=3** (3 honest topology SKIPs: oom_score_adj Linux-only, destructive- gated, nezha-required).
3. ✓ Mistborn meta-test — 43 CAUGHT / 2 ESCAPED (pre-existing P5-M20+P5-M21 from v1.0.9) / 8 SKIPPED.
4. ✓ Mistborn go test ./cmd/workable-items/... -count=3 — 30 PASS / 0 FAIL.
5. 🕒 commit_all.sh push (this commit).
6. 🕒 Nezha SSH (~ / nezha script) → cd ~/Projects/tmux && git fetch --all && git pull && bash scripts/setup.sh --rebuild && bash scripts/setup.sh --verify-only — full gate capture expected. T5 honestly SKIPs on headless Linux per §11.4.3 — T2/T3/T4 still PROVE the binding chain.

7. 🕒 Release tag v1.0.15 via GitHub + GitLab CLIs with comprehensive docs/ `changeLogs/v1.0.15.md` referenced.
8. 🕒 Post-release version bump (per operator mandate "After releasing... increase version (version code) and commit and push all work").

§3.15 — v1.0.14 clipboard copy-OUT physical proof + e2e stale-prereq fix + multi-host deploy (2026-05-22)

Status: COMPLETE (2026-05-22).

Triggering event: operator mandate (verbatim, 2026-05-22):

"We can always copy / paste from and to the terminal window and current tmux (tmx) session! Using mouse or keyboard MUST WORK properly!!! Scrolling the content / history MUST NOT be broken or anything else! After all changes are done, covered with full validation and verification tests make sure we setup new version to current host (macOS) and nezha (nezha.local is back online). Once both hosts are updated with the latest versions and on both hosts all tests are executed with success with no failure and bringing real proofs of everything working as requested and expected (physical proofs) with no bluffs commit and push all Submodules and main repo to all upstreams and release new version with comprehensive in-depth version (change) log using GitHub and GitLab CLIs."

- **A35 clipboard copy-OUT physical proof** (Fixed.md A35): identified the §101 PASS-bluff hole — bindings existed and grepped fine, but no test ever read back the OS clipboard.
 - NEW `scripts/tests/44_clipboard_copy_out_physical.sh` — operator-path. Spawns `tmx new -s NAME -d`, prints a unique marker, enters copy-mode, search-backward + select-line, invokes `@clip` (T3 direct -X + T4 the literal y keystroke that triggers the bind table end-to-end), then reads the OS-native paste tool (T5 — `pbpaste / wl-paste -n / xclip -o / termux-clipboard-get`) and asserts the marker is there. T5 SKIPS honestly with reason on headless Linux; T3/T4 binding- chain proof runs everywhere so the test is never inert. Pre-test save + post-test restore of the operator's clipboard.
 - PASS=7/0/0 this cycle (Darwin); T5 returned the marker via `pbpaste` — physical end-user evidence.
 - Layer-1 `verify.sh` static gate extended (4 new `_l1` checks for `@clip` + the three bindings).
 - Layer-3: `TMUX-CH-44` in `scripts/challenges/tmux.yaml`.
 - Layer-4: `M44` in `meta_test_false_positive_proof.sh` strips the `@clip` definition line — test 44 T1 catches universally (structural grep), T5 additionally catches where a clipboard tool is reachable. CAUGHT + RESTORED both directions.
- **A36 e2e stale podman-machine prerequisite** (Fixed.md A36): `scripts/test_e2e.sh` T1.2 still hard-required a running podman machine, a leftover from the pre-v1.0.7 SSH-bridge architecture. Fixed by probing for the native Mach-O binary FIRST and only falling back to the podman check for bridge-era installs. e2e GREEN after the fix.
- **B3 P5 escape transparency** (Issues.md B3): meta-test reported 2 ESCAPES on Mistborn — P5-M20 (non-TTY guard) and P5-M21 (cwd-capture hook). Nezha additionally reports M22 as an ENVIRONMENTAL escape (CodeGraph state). All three pre-exist this cycle. Investigation: P5-M20/M21 are layer-4 test-

DESIGN gaps — the assertions cannot distinguish "guard fired" from "fallback fired", so a single-layer strip slips through. M22 is an environmental state-dependency on the CodeGraph index/config on nezha. Underlying features GREEN. Surfaced transparently in Issues.md B3 + the v1.0.14 CHANGELOG — anti-bluff disclosure per §11.4 / §101 / §11.4.6. Closure conditions documented; deferred to a future cycle.

- **Multi-host deploy** (this turn's operator ask, satisfied): Mistborn updated to v1.0.14 via `setup.sh --rebuild`. Nezha fetched gitlab/main, fast-forwarded to v1.0.14, rebuilt and installed. Both hosts run the same code; both gates GREEN; both e2e GREEN; clipboard physically proven on Mistborn (pbpaste) and binding-chain proven on nezha (tmux buffer, T5 honest-SKIP on headless).

§3.14 — v1.0.8 uniform tmux UI recolouring (active border + clock + window-status-current) (2026-05-21)

Status: COMPLETE (2026-05-21T21:00Z).

Operator-driven follow-up to v1.0.7: "flying animated top decoration"

- clarification "Do coloring of all UI tmux parts with proper color we use instead of default green. Anything colored with that green colors has to become the color we have assigned to the bottom view we are coloring."

Pre-v1.0.8 `_apply_host_color()` set only `status-style bg=$color`. Other default-green tmux surfaces (active pane border, clock face, selected-window highlight) stayed green. v1.0.8 extends the wrapper to apply the hostname-derived colour atomically to all four:

- `status-style` `bg=$color`
- `pane-active-border-style` `fg=$color`
- `clock-mode-colour` `$color`
- `window-status-current-style` `bg=$color,fg=black`

`mode-style` + `message-style` deliberately untouched (default yellow, not green; yellow provides best contrast against any palette bg).

NEW test 26 (26_ui_color_uniformity.sh) — operator-path session spawn + live show `-gv` readback per surface — PASS=5/0/0 on Darwin. NEW M24 paired mutation — regex-strips the three v1.0.8 set-lines from the generated wrapper, asserts test 26 T2/T3/T4 FAILs. MUTATION CAUGHT + FEATURE INTACT both directions.

Captured operator-path readback (Mistborn, this session):

- `status-style: bg=colour44`
- `pane-active-border-style: fg=colour44`
- `clock-mode-colour: colour44`
- `window-status-current-style: bg=colour44, fg=black` → All four surfaces uniformly turquoise (colour44 = RGB 0,215,215).

Verification (Darwin quintuple-fresh, 2026-05-21):

- `setup.sh --verify-only` → PASS=24 FAIL=0 SKIP=2
- `meta-test` → 36 caught / 0 escaped / 6 skipped

- e2e → PASS=9 FAIL=0 SKIP=0
- codegraph_validate → PASS=4 FAIL=0 SKIP=1

Released v1.0.8; setup.sh re-run as final install per operator "Make sure we setup the new version once all done." directive.

§3.13 — v1.0.7 cross-host portability + hostname-colour rebalance + Node-22 pin + release (2026-05-21)

Status: COMPLETE (2026-05-21T18:30Z).

Operator-driven cycle: tried to install on Nezha (Linux), 3 tests FAILED → 6 fix rounds → Nezha GREEN → operator reported orange-color collision (Nezha + Mistborn) → palette rebalanced + test 25 + M23 mutation → release v1.0.7.

Six fix rounds (all Nezha-driven; live forensic capture):

1. Initial 3-fix attempt (test 09 timing, test 17 ingestion race, test 21 codegraph bootstrap). Test 17 fixed; tests 09 + 21 had secondary defects.
2. Bash -lt fractional bug in round-1 poll-loop (silent fail); npm-prefix PATH augmentation for non-interactive shells.
3. CodeGraph init clobbers config.json on every version; auto-update wiring per §11.4.80 mandate ("ALWAYS latest").
4. cgroup.procs sweep (kill all PIDs not just first) + stat -f '%z' portability bug (Darwin vs Linux).
5. Test 09 T4.2 rewritten to verify the REAL containment invariant (cgroup.procs drained), not systemd's unit-state transition (which is sticky on systemd 258 / ALT 11 — verified live).
6. macOS Node 22 LTS pin: codegraph 0.8.0 refuses Node 25 per upstream issue #81. Operator authorised brew install node@22 && brew link --force --overwrite. Updated .zshrc references.

Then the hostname-colour palette rebalance:

- Operator: "nezha and Mistborn both show orange". Palette had 7 orange-family colours out of 27.
- Rebalanced to 27 entries spanning the hue spectrum; no two adjacent within RGB Euclidean distance 80.
- NEW test 25: T1 nezha+Mistborn distance ≥ 80 (post-fix: 332.7); T2 16-synthetic-hostname pairwise within pigeonhole tolerance; T3 palette adjacency check.
- NEW M23 mutation: revert palette → test 25 FAILs.

Verification (Darwin quintuple-fresh, 2026-05-21):

- setup.sh --verify-only → PASS=23 FAIL=0 SKIP=2
- meta-test → 34 caught / 0 escaped / 6 skipped
- e2e → PASS=9 FAIL=0 SKIP=0
- codegraph_validate → PASS=4 FAIL=0 SKIP=1
- codegraph_cadence_check → GREEN

Nezha verification: round-5 setup.sh full pipeline GREEN (after fix rounds 1-5 landed). v1.0.7 includes all 6 rounds + palette + test 25.

Out-of-scope this cycle (logged honestly per §11.4.6):

- Auto-detection of Node version compatibility in `setup.sh` — deferred to v1.0.8.
- Pre-tag Nezha re-verify of v1.0.7 final → will run after tag push as deferred post-release smoke.

§3.12 — v1.0.6 workable-items closure + §11.4.80 cadence + Containers/QWEN.md + release tag (2026-05-21)

Status: COMPLETE (2026-05-21T15:30Z).

Operator directive (verbatim): "Finish all workable items you have presented here, rebuild and install again, retest fully and if fully working as expected release new version!"

Closed (this cycle):

- **§11.4.80 automatic-trigger wiring** (was the one deferred item from v1.0.5 audit). New `scripts/codegraph_cadence_check.sh` + `scripts/codegraph_install_cadence.sh`. Darwin launchd plist installed + loaded (verified via `launchctl list`). Cross-platform git pre-push hook installed (warns on STALE; `CADENCE_MODE=block` available). Anti-bluff: cadence check reads STAMP CONTENT (regenerated `_at` + `_count`), not just existence. §11.4.81 cross-platform-parity dispatch in the installer: Linux gets systemd user timer; Darwin gets launchd plist.
- **Containers/QWEN.md covenant gap** (was the second-to-last out-of-scope item from v1.0.5 — Containers/QWEN.md create). Containers commit `fbef9d6` pushed to github + gitlab. Parent pointer bumped `4ca5491` → `fbef9d6` in this cycle's commit.
- **Full rebuild + install** via `bash scripts/setup.sh` (not just `--verify-only`). `~/.tmux.conf` installed; `.bashrc` + `.zshrc` snippets appended; tmx wrapper installed.
- **Quintuple-fresh verification** captured this session: `setup.sh --verify-only` → `PASS=22/0/SKIP=2`; `meta-test` → `32 caught / 0 escaped / 6 skipped`; `e2e` → `9/0/0`; `codegraph_validate` → `4/0/SKIP=1`; `codegraph_cadence_check` → GREEN.

Still out of scope (logged honestly per §11.4.6):

- Cadence-script paired §1.1 mutation (backdate-stamp → cadence FAILs) — deferred to v1.0.7. The script's content-based read is already anti-bluff; the mutation would tighten regression protection.
- CodeGraph upstream `--include-submodules` — separate cycle per §11.4.74.
- Linux-host CI runner — would exercise the Linux systemd user timer install path + the Linux branches of tests 09/13/14.

Release v1.0.6 tagged + pushed per §11.4.40 (complete fresh retest preceded the tag; no spot-check shortcut).

§3.11 — §11.4.81 cross-platform-parity + Darwin branches + constitution §11.4.81 (2026-05-21)

Status: COMPLETE (2026-05-21T07:30Z).

Triggering events (in arrival order):

- Operator (2026-05-21): "Any Linux-only blocker / issue we have MUST BE created macOS and other supported platforms equivalent! So, depending on platform proper implementation will be used for particular OS! EVERYTHING MUST BE PROPERLY EXTENDED AND UPDATED!"
- Operator (2026-05-21): "Add this OS / Platform details / rules / mandatory constraints into our root (constitution Submodule) Constitution.md, CLAUDE.md, AGENTS.md, QWEN.md and other constitution Submodule relevant files so all projects who inherit these in the future immediately in such situations create proper missing implementations for particular platform / OS! Make sure you first fetch and pull the latest version of constitution Submodule. Once all done and extended properly commit and push constitution Submodule to all upstreams, then re-process and re-evaluate all rules and mandatory constraints..."
- (Power blackout mid-cycle — work resumed cleanly from working tree.)

Plan + execution captured at docs/plans/v1.0.5.md. Seven phases all landed in one v1.0.5 commit per §11.4.42 iteration discipline.

- **§11.4.26 step 1+:** constitution submodule fetched + ff-merged from 7f738df → 19ce1b1 (brought in §11.4.79 + §11.4.80 CodeGraph anchors).
- **§11.4.81 anchor LANDED in constitution submodule (pushed 6e164f3 to origin HelixDevelopment).** Universal cross-platform- parity mandate + mirror blocks in Constitution.md / CLAUDE.md / AGENTS.md / QWEN.md. Three sub-mandates: (A) per-OS implementation REQUIRED via runtime dispatch; (B) per-OS tests REQUIRED with positive captured evidence per branch; (C) honest kernel-gap citation + adjacent equivalent test REQUIRED where no equivalent exists. constitution/QWEN.md ALSO gained the verbatim 2026-04-28 anti-bluff covenant quote (audit gap fix).
- **A26 — §11.4.79 compliance fix.** Removed constitution/** and Containers/** from .codegraph/config.json exclude (own-org MUST be INCLUDED); kept tmux/** excluded (third-party). scripts/codegraph_validate.sh NEW — 5 probes (V1 CLI version, V2 node count, V3 §11.4.79 split, V4 honest-gap re submodule traversal, V5 MCP spawn). M22 paired mutation (re-exclude → V3 FAILs). Honest gap: CodeGraph 0.6.8 doesn't traverse submodules; config compliance met but practical cross-submodule indexing waits for upstream CodeGraph support.
- **A25 + A22 + A24 — Darwin branches for tests 09/13/14 + NEW test 24 (per §11.4.81 just landed).** All four exercise the macOS POSIX rlimit primitives that are the kernel-enforced equivalents of the Linux cgroup primitives:
 - **Test 09 Darwin (PASS=6/0/0):** wrapper invokes tmx-rlimit-wrapper.sh, two operator-path sessions, distinct server PIDs, ulimit -t/-u readback inside each pane via send-keys + capture-pane, SIGKILL session A's server, session B survives with ORIGINAL PID (positive evidence per §11.4.5).
 - **Test 13 Darwin (PASS=2/0/0):** child bash lowers ulimit -u 64
 - fork-bombs; captures EAGAIN occurrences from stderr (bash: fork: Resource temporarily unavailable)=XNU kernel-enforced RLIMIT_NPROC.

- **Test 14 Darwin (PASS=5/0/0):** 3 operator-path sessions; SIGKILL session A's server (macOS adjacent test for OOM-independence per §11.4.81 (C) — Darwin has no OOM killer); verify B+C survive with ORIGINAL PIDs + tmx ls still lists them.
- **NEW Test 24 Darwin (PASS=2/0/0):** child bash sets `ulimit -t 2`
 - CPU-bound loop; verifies process killed by signal 24 (SIGXCPU) after ~3s wall (exit rc=152 = 128+24); also verifies `TMX_CPU_HARD_SEC=7200` propagates to `RLIMIT_CPU=7200` inside session. §11.4.81 (C) adjacent test for the XNU `RLIMIT_AS` gap.
- **A25 paired mutations refresh.** M7-M10 RETIRED (targeted dead scripts/ `tmx-vm` VM-wrapper path, replaced by native dual-OS per Fixed.md A4-A8). M20 + M21 NEW for Darwin `rlimit`. M22 for §11.4.79 codegraph exclude. Meta-test on Darwin: 32 caught / 0 escaped / 6 skipped (M4/M5 topology-correct SKIP on Darwin; M7-M10 retired-with-rationale SKIPS).
- **Constitution submodule pointer bumped** 19ce1b1 → 6e164f3 in this same project commit per §11.4.26 step 7.

Full gate this cycle (Darwin arm64, 2026-05-21): `setup.sh --verify-only`
 PASS=22/0/SKIP=2, meta-test 32/0/SKIP=6 (all SKIPS §11.4.3 topology-correct or retired-with-rationale), e2e 9/0/0, codegraph_validate PASS=4/0/SKIP=1.

Out-of-scope this cycle (honest tracking per §11.4.6):

- §11.4.80 automatic-trigger wiring (cron / git hook for the constitution-provided `codegraph_update.sh` + `codegraph_sync.sh`) — deferred. Manual invocation works today.
- CodeGraph upstream `--include-submodules` support — out of scope per §11.4.74.
- Containers/QWEN.md create — separate PR to vasic-digital/Containers per §11.4.28.
- Linux-host CI runner — would let us exercise the Linux branches of tests 09/13/14 alongside the Darwin branches running today.

§3.10 — CodeGraph (§11.4.78) + verbatim covenant + AUDIT fixes + docs reorg (2026-05-21)

Status: COMPLETE (2026-05-21T05:30Z).

Triggering events (in arrival order):

- Operator (2026-05-21): create full working plan for tackling all these points completely; all existing tests + Challenges MUST work anti-bluff; covenant MUST be part of project + submodule Constitution / CLAUDE / AGENTS; HelixConstitution submodule is the root.
- Operator (2026-05-21): incorporate / install CodeGraph for Claude Code, OpenCode, Kimi CLI, Crush, Qwen Code; comprehensive anti-bluff tests.
- Operator (2026-05-21): full documentation + user guides + HTML + PDF exports.
- Operator (2026-05-21): do NOT modify constitution submodule, only keep it regularly updated.
- Operator (2026-05-21): documentation under context-named subdirectories per the constitution rule.

Plan + execution captured at docs/plans/v1.0.4.md. Eight phases all landed in one v1.0.4 commit per §11.4.42 iteration discipline.

- **A18 CodeGraph integration (§11.4.78).** CLI v0.6.8 installed; .codegraph/config.json tracked with §11.4.10 secret exclusions + §11.4.28 owned-submodule paths; .codegraph/codegraph.db gitignored; §11.4.77 regen manifest + scripts/codegraph_reindex.sh. MCP wired for all 5 agents (Claude Code .mcp.json NEW; OpenCode + Kimi audited; Crush .crush.json NEW; Qwen Code .qwen/settings.json NEW). All configs reference bare codegraph on PATH. Comprehensive docs/codegraph/README.md. Tests 20/21/22, challenges CH-20/21/22, mutations M16/M17.
- **A19 verbatim covenant propagation.** §11.4 2026-04-28 user mandate now LITERALLY present in project CLAUDE.md + AGENTS.md + QWEN.md (was only in Constitution.md). Verify.sh Layer-1 gate added; test 19 PASS=7/0/0; CH-19; M15 (mutates a temp copy — real CLAUDE.md is never touched).
- **A20 AUDIT-1 fix.** M4/M5 meta-test mutations were silently SKIPping on Darwin for the wrong reason (BSD sed -i quirk). Real root cause: the mutations target Linux-only cgroup/systemd-run wrapper code unreachable on Darwin. Fix: explicit uname -s topology guard per §11.4.3 + portable inplace_sed for Linux runs.
- **A21 AUDIT-2 fix.** tmx kill -t NAME (documented friendly verb) was ambiguous because tmux only knows kill-pane/kill-server/ kill-session/kill-window. Wrapper now translates the bare kill verb to kill-session. Test 23 PASS=5/0/0; CH-23; M19.
- **Docs reorganised** under context-named subdirectories per the operator's invocation of the constitution rule: 7 docs moved (GUIDE/SCROLLING/CODEGRAPH/CONTAINERIZATION_PLAN/NATIVE_DUAL_OS_PLAN/PER_SESSION_ISOLATION_PLAN/PLAN_v1.0.4). Every reference in governance + scripts updated atomically.
- **§11.4.65 universal-Markdown export.** New scripts/export_docs.sh (pandoc HTML + weasyprint PDF, idempotent, per-file timeout 60s) refreshed siblings for every consumer Markdown.
- **§11.4.71 pre-push.** Parent + constitution/ + Containers/ all at upstream tip; no divergent commits.
- **Honest gaps logged (§11.4.6) — out of scope this cycle:**
 - Upstream constitution/QWEN.md covenant insert → separate PR to HelixDevelopment/HelixConstitution (operator forbids modifying constitution from inside this project).
 - Containers/QWEN.md create + populate → separate PR to vasic-digital/Containers per §11.4.28.
 - Shell parser for CodeGraph (currently only the C file indexed, 6 nodes) → upstream contribution to add tree-sitter shell.
 - Agent-driven unforgeable-challenge end-to-end test → classified AUTONOMOUS_DEIGNED per §11.4.52 carve-out (mechanical seam exists via test 22 T7; agent-driven layer lands when a headless agent harness is wired).

Full gate this cycle (Darwin arm64, 2026-05-21): `setup.sh --verify-only`
PASS=18/0/SKIP=5, meta-test 26/0/SKIP=6 (all SKIPS §11.4.3 topology-correct),
e2e 9/0/0.

§3.9 — Scrolling fix + HelixConstitution inheritance (2026-05-21)

Status: COMPLETE (2026-05-21).

Triggering event: operator research note — tmux scrolling (up/down of terminal output, especially inside the Claude Code TUI) did not work; requirement to scroll vertically from ANY computer or mobile phone (Termux/Android). Plus the mandate to incorporate the HelixConstitution submodule as the root of the Constitution / CLAUDE.md / AGENTS.md, inherited further.

- **A16 scrolling fix** (Fixed.md A16): `scripts/tmux.conf.template` — `history-limit 2000→50000`, `mode-keys vi`, `WheelUp/WheelDown` overridden to always drive copy-mode scrollbar (works on desktop mouse + trackpad + Termux touch-scroll), `allow-passthrough + extended-keys + terminal-features` `extkeys` for the Claude Code TUI, OS-adaptive `@clip` clipboard. NEW test 17 (operator-path, PASS=13/0/0), `verify.sh` Layer-1 static gate, TMUX-CH-17, M12 + M13 paired mutations.
- **A17 HelixConstitution** (Fixed.md A17): added the `constitution/` submodule (pinned 7f738df). Full refactor of `Constitution.md` to the `extends-template` form (Project Articles §101–§109); `CLAUDE.md` / `AGENTS.md` inheritance pointers; new `QWEN.md`. NEW test 18 (PASS=10/0/0), TMUX-CH-18, M14 + CM-CONSTITUTION-INHERITANCE paired mutations. The constitution mutation runs on a TEMP COPY — the real, decoupled `constitution/` submodule is never modified (operator directive).
- **Containers submodule**: adopted remote 4ca5491, which already carries HelixConstitution recursive inheritance (`find_constitution.sh`, `QWEN.md`, all four governance docs). Parent gitlink bumped b077f2c → 4ca5491.
- **Meta-test portability**: M1/M2/M3/M6 converted from GNU `sed -i` to a portable `inplace_sed` helper — they now run on macOS instead of silently skipping. Meta-test: 18 caught / 0 escaped / 6 skipped (was 10/0/10).
- Full gate this cycle (Darwin arm64, 2026-05-21): `setup.sh --rebuild`
GREEN, suite PASS=13/0/SKIP=5, meta-test 18/0/6, e2e 9/0/0.

§4 — Recent commits

- (this commit, 2026-05-21) — **A16 scrolling fix + A17 HelixConstitution inheritance (v1.0.3 / versionCode 4)**. `tmux.conf.template`: `history-limit 50000`, `vi` copy-mode, `WheelUp/Down` → copy-mode scrollbar override, Claude Code passthrough, OS-adaptive clipboard. HelixConstitution submodule added at `constitution/`; `Constitution.md` refactored to the `extends-template` form (§101–§109); `CLAUDE/AGENTS` inheritance pointers + new `QWEN.md`. Tests 17 + 18, challenges CH-17/18, mutations M12–M14 + CM-CONSTITUTION-INHERITANCE; meta-test M1/M2/M3/M6 made portable. Containers gitlink → 4ca5491. Gate: PASS=13/0/5, meta 18/0/6, e2e 9/0/0. See Fixed.md A16 + A17, §3.9.
- (this commit, 2026-05-16) — **A15 cosmetic .exe-strip cycle (v1.0.2 / versionCode 3)**. Bottom-left status bar showed `claude.exe` because Claude Code v2.x ships its macOS native binary literally as `.../@anthropic-ai/`

- claude-code/bin/claude.exe (verified Mach-O 64-bit ARM64). Patched scripts/tmux.conf.template to set automatic-rename-format "{s/\.exe\$//:pane_current_command}" — literal-dot-anchored, escape-verified-empirically. Added test 16 (operator-path per §11.4.7, includes regression guard binary t16_bashehex to catch the unescaped-dot bug class), challenge TMUX-CH-16, M11 paired mutation. Live operator-path validation recorded: pane_current_command='claude.exe' → #W='claude'. Full gate: PASS=11 FAIL=0 SKIP=5 (same SKIP profile as v1.0.0). See Fixed.md A15.
- (commit before that, 2026-05-13) — Audit cycle: tmux submodule reverted from 3f651d9f to cc117b5 (3.6a tag) after silent pin-drift caught; README.md test-count + PASS/SKIP staleness fixed (8→14 tests, PASS=6→10, SKIP=2→4); CLAUDE.md + AGENTS.md gained project summary, cross-links, fresh-conversation workflow, NN_*.sh glob fix, named four layers, "Files to never edit directly"; Issues.md restored A/B/C/D/E section headers; CONTINUATION.md §3.8 + timestamp refresh per §12.10. See §3.8.
 - f4132aa — Auto-commit: opaque submodule bumps (Containers e377dea→af51968 legitimate upstream governance; tmux cc117b5→3f651d9f reverted in this commit). §12.10 / §11.4.6 violation documented in §3.8.
 - (previous commit) — §11.4.4 layer-4 paired-mutation harness (META-MUT-001): meta_test_false_positive_proof.sh with 5 mutations (10/0/0 all caught); CLAUDE.md/AGENTS.md PENDING→LANDED; Issues.md now empty of open items. Also: fixed 5 broken script references (REPO_ROOT wrong depth, filenames); implemented systemd-run --user --scope wrapping in tmx.template; test 09 topology dispatch fixed (functional probe, not mount grep); CLAUDE.md compacted to match AGENTS.md; binary built and verified GREEN (PASS=10 FAIL=0 SKIP=4: OOM helper + 3 destructive).
 - 39284ab — Full coverage: CH-09 added (was missing); tests 12/13/14 (T5/T7/T8 destructive) with TMX_TEST_DESTRUCTIVE=1 gate; topology probe in tmx.template; challenge entries CH-12/13/14; B1/C1/C2/C3/D1 migrated Issues→Fixed.
 - 8b37046 — Hostname-derived status-bar colour: DJB2→27-colour palette algorithm + wrapper integration + tests 10/11 + challenges + docs.
 - 68a65b0 — Anti-bluff enforcement: TEST-AUDIT-001 complete (9 tests §11.4.2-audited), challenges paths fixed, AGENTS.md compact rewrite, covenant propagation verified.
 - b92bf7f (2026-05-08) — §1 anti-bluff covenant verbatim user-mandate quote added; test 09 crash-isolation-scope (14/0/0) landed.
 - 08d4ba5 (2026-05-07) — Initial vasic-digital/tmux: migrated from ATMOSphere project + per-session containerization plan + 8-test verification gate.

§8 — Resumption recipe

1. cd ~/Projects/tmux
2. Read this document
3. Read Constitution.md, CLAUDE.md, AGENTS.md, Issues.md, Fixed.md
4. Find the topmost IN PROGRESS / OPEN item, resume
5. Update this document in the SAME commit as the work itself (§12.10 invariant)